Pratyush Bhatt

State College, PA | pvb5440@psu.edu | https://www.linkedin.com/in/pratyushbhatt–/ | https://pratyushbhattrobotics.github.io/

Education

The Pennsylvania State University — State College, PA
Bachelor of Science in Mechanical Engineering — *Expected Graduation May 2027*

Experience

Research Assistant – Robotics Lab — October 2025 – Present
The Pennsylvania State University — *State College, PA*

- Built 2 hexapod robots for locomotion experiments, by assembling mechanical components, making structural adjustments, and implementing gait patterns for physical testing.
- Conducted 500 experimental trials across different phase shifts and duty cycles, comparing different gait patterns to evaluate locomotion efficiency.
- Collected and analyzed motion data from 500 trials using MATLAB and Vicon Nexus, assessing how leg timing and ground-contact duration affected gait performance.

Mechanical Engineering Intern — June 2026 – July 2026
Baxter International — *Batesville, IN*

- Designed and developed a modular experimental test apparatus for testing airflow through porous materials, reducing air leakage from 35% to less than 10% by iterating CAD models in solid edge.
- Evaluated the test fixture under a 600 lb clamping load, driving more reliable structural design decisions, by analyzing load distribution and clamp placement using Finite Element Analysis (FEA)
- Tested the apparatus with 4 different porous materials, by assessing its suitability for airflow testing across different material types.

Course Grader – Computational tools — January 2026 – Present
The Pennsylvania State University — *State College, PA*

- Evaluating SolidWorks modeling and simulation assignments for 200 students, by assessing CAD models and FEA results.

Projects

Vision-Guided Robotic Arm | *Embedded Systems* — August 2026 – October 2026

- Built a 3-DOF vision-guided robotic arm, leading to autonomous pick-and-place operation, by integrating mechanical components, OpenCV object detection, and motor control.
- Achieved 35 successful pick-and-place cycles across 40 physical trials, driving an 87.5% success rate, by converting detected object locations into arm movements and calibrating joint limits.
- Configured the arm in ROS 2, leading to a setup for visualizing and testing planned movements, by using RViz for visualization, Gazebo for simulation, and MoveIt 2 for motion planning.

Self-Balancing Robot | *Feedback control* — September 2026 – October 2026

- Built a 2-wheel self-balancing robot that balanced successfully in 8 of 10 trials, leading to an 80% success rate, by integrating an MPU6050 sensor, an ATmega328P microcontroller, and motor control.
- Implemented PD feedback control in C++, maintaining tilt within 10 degrees of upright, by combining accelerometer and gyroscope readings and tuning control gains.
- Added motor shutdown at a 25° tilt, leading to automatic stopping when the robot tipped beyond its operating limit, by monitoring the measured angle.

Handheld Visual SLAM

- Developing a handheld SLAM system for a 120 sq ft indoor space, aiming to track camera movement and map the surroundings, by processing camera images using ROS 2 and OpenCV.

BattleBots Robotics Competition | *Mechanical Design* — October 2025 – January 2026

- Designed and fabricated robot chassis within a 6 lb total robot weight limit, supporting drivetrain and weapon mounting, by creating CAD models in Autodesk Fusion 360 and assembling the frame.
- Collaborated with mechanical, electrical, and controls teammates to integrate the drivetrain and weapon systems, enabling coordinated robot operation by aligning mounting locations and checking component fit.

Technical Skills

CAD: SolidWorks, Solid Edge, Autodesk Fusion 360, AutoCAD
Analysis and Simulation: ANSYS, MATLAB, Vicon Nexus
Robotics and Embedded Systems: ROS 2, MoveIt 2, OpenCV, Gazebo, RViz
Programming: Python, MATLAB, C++